GATE Computer Science – Sample PYQ Practice Set

Synthetic practice document created for testing an exam-preparation application's PDF upload and ingestion pipeline. These are sample questions, not an official GATE question paper.

Q1. DBMS – Normalization

Consider a relation R(A, B, C) with functional dependencies A $\rightarrow$ B and B $\rightarrow$ C. Which statement is correct?

A. A $\rightarrow$ C can be inferred

B. C $\rightarrow$ A can be inferred

C. B is a candidate key

D. The relation is necessarily in BCNF

Q2. Operating Systems – Scheduling

Three processes P1, P2 and P3 arrive at time 0 with CPU burst times 6 ms, 2 ms and 4 ms respectively. Under non-preemptive SJF, which process executes first and what is the total waiting time?

Q3. Algorithms – Complexity

What is the worst-case time complexity of searching for an element in a balanced binary search tree containing n nodes?

Q4. Theory of Computation

Which of the following languages is guaranteed to be regular?

A. A language recognized by a finite automaton

B. A language recognized only by a pushdown automaton

C. Every context-free language

D. Every recursively enumerable language

Q5. Computer Networks

Explain the purpose of flow control and distinguish it from congestion control in a computer network.

Q6. Data Structures

A max-heap contains n elements. State the time complexity of inserting a new element and explain why.

Q7. Digital Logic

Simplify the Boolean expression F = A'B + AB + AB'. What is the resulting expression?

Q8. Computer Organization

What is the primary purpose of a cache memory? Explain how locality of reference affects cache performance.

Q9. Compiler Design

Identify the major phases of a compiler from lexical analysis through code generation.

Q10. Probability

A fair coin is tossed three times. What is the probability of obtaining exactly two heads?

Answer Key / Reference Concepts

Q1: A. By transitivity, $A \rightarrow B$ and $B \rightarrow C$ imply $A \rightarrow C$.

Q2: P2 executes first. With order P2, P3, P1, waiting times are 0, 2, and 6 ms; total = 8 ms.

Q3: O(log n).

Q4: A. Languages recognized by finite automata are regular.

Q5: Flow control prevents a fast sender from overwhelming a receiver; congestion control addresses overload within the network.

Q6: O(log n), because the inserted element may move up the heap height.

Q7: A + B.

Q8: Cache reduces average memory-access time by exploiting temporal and spatial locality.

Q9: Lexical analysis, syntax analysis, semantic analysis, intermediate representation/code generation, optimization, and target code generation.

Q10: 3/8.